Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

(82)

Mid Semester Examination

March 16-03-2018

Max. Marks: 30 | Duration: 1Hrs30Min
Class: T.E. | Semester: VI
Course Code: ETC605 | Branch:Elect. and Telecom
Name of the Course: Operating System

Instruction:

(1) All questions are compulsary
(2) Draw neat diagrams
(3) Assume suitable data if necessary

<table>
<tr><th>Q No.</th><th></th><th>Max. Marks</th><th>CO</th></tr>
<tr><td>Q.1</td><td>Consider the following set of processes,with the length of the CPU burst given in milliseconds:
<table><tr><td>Process</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td></tr><tr><td>Burst Time</td><td>10</td><td>1</td><td>2</td><td>1</td><td>5</td></tr><tr><td>Priority</td><td>3</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table>
The processes are assumed to have arrived in the order P1,P2,P3,P4,P5, all at time0.
(a) Draw four Gantt charts that illustrates the execution of these processes using the following scheduling algorithems: FCFS, SJF, nonpreemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1).
(b) What is the turnaround time of each process for each of the scheduling algorithems in part a?
(c) What is the waiting time of each process for each of these scheduling algorithms?
(d) Which of the algorithms results in the minimum average waiting time (over all processes)?</td><td>10</td><td>CO2</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
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solve above questions for this tabular data.</td><td>10</td><td>CO2</td></tr>
<tr><td>Q.2</td><td>Justify the necessity of each of the five process states through process state transition diagram and also list transition activities.</td><td>05</td><td>CO2</td></tr>
<tr><td>Q.3</td><td>Discuss USER and SYSTEM services provided by an operating system.</td><td>05</td><td>CO1</td></tr>
<tr><td>Q.4</td><td>What are the requirement of RTOS?</td><td>05</td><td>CO1</td></tr>
<tr><td>Q.5</td><td>Compare desktop OS versus Embedded OS.</td><td>05</td><td>CO1</td></tr>
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